

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

|                                                   |                             |
|---------------------------------------------------|-----------------------------|
| Mid-Transit Time (Tmid)                           | 2459979.59 +/- 0.14 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.153 +/- 0.047             |
| Transit depth (Rp/Rs)^2                           | 2.35 +/- 1.44 [%]           |
| Orbital Inclination (inc)                         | 87.88 +/- 2.89              |
| Airmass coefficient 1 (a1)                        | 10.02 +/- 6.37              |
| Airmass coefficient 2 (a2)                        | -1.17 +/- 0.62              |
| Scatter in the residuals of the lightcurve fit is | 61.27 %                     |
| Best Comparison Star                              | None                        |
| Optimal Aperture                                  | 1.62                        |
| Optimal Annulus                                   | 5.13                        |
| Transit Duration (day)                            | 0.033 +/- 0.04              |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R fit vs published     | 0.153 ± 0.047 vs 0.1213 (deviation 0.67σ) |
| Duration fit vs published | 0.033 d vs 0.1033 d (deviation 1.76σ)     |
| Mid-time O-C              | -108.1 min                                |
| Depth SNR                 | 3.3                                       |
| Residual scatter          | 61.27 %                                   |

Light curve

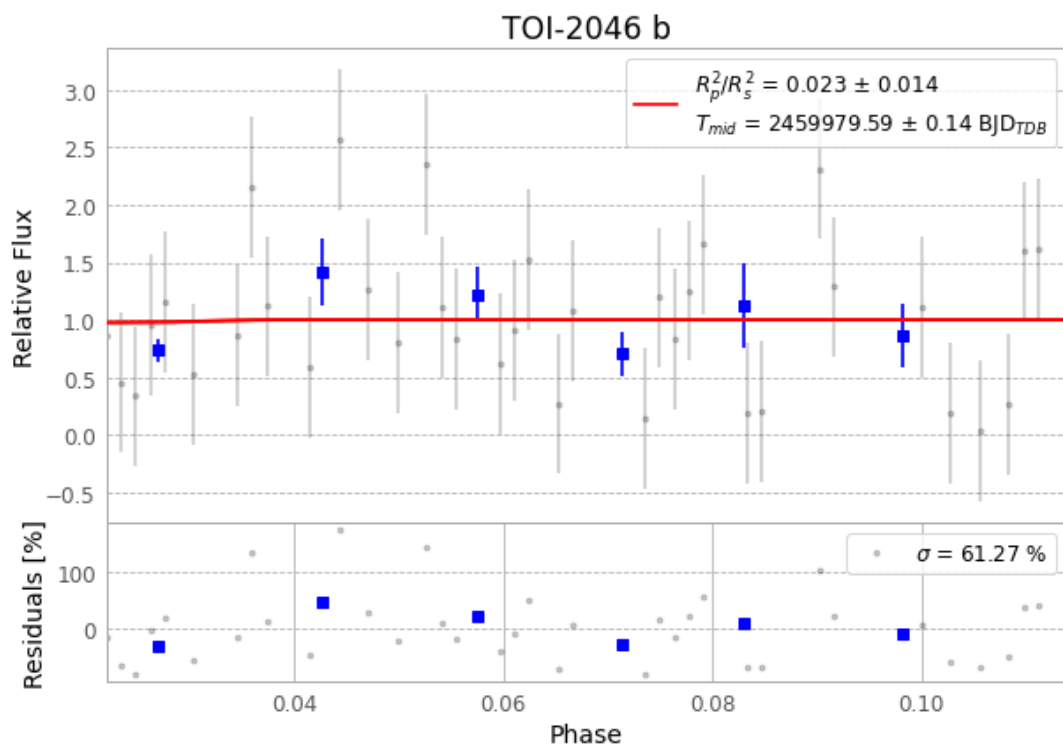


Figure 1 — FinalLightCurve\_TOI-2046 b\_2023-02-03.png (SHA-256 a2ea3459c1650fba...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 392], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 7b1876c7f5859455...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

70 FITS frames (46 MB), TOI2046230204024515.FITS → TOI2046230204061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-230204.log (f52653f43ade...), FinalLightCurve\_TOI-2046 b\_2023-02-03.png (a2ea3459c165...), FinalLightCurve\_TOI-2046 b\_2023-02-03.csv (59301c840f1e...)

## Compute resources

Wall time 172.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:03Z.